

Expert Opinion

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Advances in the management of organophosphate poisoning

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Organophosphate (OP) poisoning is commonly encountered in agricultural communities. The mainstay of therapy in OP poisoning is the use of atropine. However, several other therapies have been evaluated. Although oxime has been the most studied antidote, results in humans have been disappointing and limited by the lack of well-designed, prospective, randomised controlled trials. The key factor in determining outcomes in OP poisoning appears to be the timing of antidote administration. Other adjuvants, such as magnesium, fresh frozen plasma and haemoperfusion appear promising, and need to be explored further. A multi-faceted approach may be the answer to improving outcomes in OP poisoning. This review evaluates the advances in OP management over the last 20 years.

Keywords: haemoperfusion, insecticide, meta-analysis, oxime, random effects

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1. Introduction

Organophosphate (OP) poisoning is common in agricultural communities [1,2] and has significant effects in developing countries, not only on the already stretched health system in terms of resource allocation, but also on the individual, who most often has to shoulder the financial burden of treatment. A large proportion of patients admitted with poisoning to intensive care units (ICUs) of developing countries have insecticide poisoning, notably from OP compounds [2,3]. This has led to the evaluation of several treatments that would effectively reverse the protean manifestations of poisoning [4], and improve outcomes. Although atropine is considered to be the specific antidote for OP poisoning, much work has gone into the assessment of the effect of nucleophilic agents (oximes) on outcomes in experimental poisoning in animals, as well as human poisoning [5]. These oximes cleave covalently bound OP off the OP-acetylcholinesterase (AChE) conjugate, thereby regenerating the AChE. This pharmacological effect of oximes has not translated, in oxime trials in human OP poisoning, to improvements in clinical outcomes [5]. Whilst the methodology of some of the oxime trials has been questioned [6], it is possible that mega-intoxications and delayed presentations may have contributed to the observed lack of benefit of oximes [5]. There have also been several advances in the understanding of the pathophysiology of OP poisoning that have led to the assessment of other agents and therapies in animal, as well as human OP poisoning. These advances have been in terms of recognition of cardiac manifestations in OP poisoning [7,8], pathophysiology of respiratory failure [9], as well as cellular mechanisms that may play a role in the progression of paralysis [10]. This review focuses on the existing evidence for the role of antidotal and adjunctive therapy in OP poisoning and specifically addresses the role of oximes in acute human ingestional and inhalational OP exposure.

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2. Review of the literature

2.1 Materials and methods

2.1.1 Selection of trials

Controlled clinical trials on acute OP poisoning in humans that evaluated specific therapies in OP poisoning were considered for evaluation. We considered for inclusion only studies that compared a particular therapy (e.g., oximes) and standard care (atropine, ventilation and other supportive care) with standard care alone. The specific therapy primarily focussed on oximes (pralidoxime or obidoxime), although other adjuncts, such as magnesium, haemoperfusion, alkalinisation and fresh frozen plasma (FFP), were also screened. Only trials reporting hospital mortality or need for invasive mechanical ventilation were included. Both retrospective trials (historical and non-randomised) and prospective trials were considered. The inclusion of retrospective trials in this review was because of a paucity of prospective controlled trials in human OP poisoning. There were no language restrictions. Electrophysiological studies and studies on animals were not included in the analysis; however, physiological basis for effectiveness of therapy was explored from these studies, as well as from experimental models.

2.1.2 Search strategy

A computerised literature search using the National Library of Medicine MEDLINE® and OVID® for the period 1966 to December 2006 was performed in order to identify articles on therapies in OP poisoning, specifically focussing on oximes. The search encompassed a range of search terms including organophosphate OR organophosphorus OR pesticide OR insecticide AND poison or poisoning (MESH) AND oxime or pralidoxime OR treatment OR therapy (MESH). In addition, further exploratory searching was done combining the terms organophosphate poisoning with magnesium, dialysis, haemoperfusion, alkalinisation AND plasma OR FFP. We also appraised review articles on OP to identify other studies on this subject. Studies identified by literature searches and other controlled trials on OP poisoning were also scrutinised to identify additional articles pertaining to therapy in OP poisoning. Personal correspondence to authors was sought to clarify details of trial, where necessary.

2.1.3 Quality assessment

This was performed on the mode of randomisation, definition of criteria for inclusion and exclusion, objective criteria for intubation and need for intensive care, as well as intention-to-treat analysis. No formal assessment of study quality was undertaken [11].

2.1.4 Statistical methods

Historical and non-randomised studies were categorised as 'retrospective' and prospective randomised studies were categorised as 'prospective' for analysis and presentation. DerSimonian and Laird [12] random effects meta-analyses were

undertaken in each of the retrospective and prospective groups where interest is in the risk difference (RD). Finally, the risk difference and variance estimates from each of the retrospective and prospective meta-analyses were combined to obtain an overall estimate of the RD. Heterogeneity of treatment effects was diagnosed using the *Q* statistic at a *p* value of ≤ 0.1 [13]. The 'rmeta' package [14] and author-created functions were employed in R [101] for performing all analyses. All reported tests were two-sided.

2.2 Results

Of the 1301 articles on OP poisoning identified by literature search, 629 articles pertained to treatment in OP poisoning and 13 trials assessed the role of oximes in human OP poisoning. This included four randomised controlled trials (RCTs) [15-18], one historical controlled study [19], five retrospective studies [20-24], a prospective trial of two [25] and three treatment arms [26], as well as a single-arm descriptive study of continuous pralidoxime administration [27]. The 8 trials comparing oxime therapy (treatment arm) with no oxime therapy (control arm) are summarised in Table 1 [15,17,19-22,25,26]. Two meta-analyses [5,28], as well as two systematic reviews [6,29] and a Cochrane review [30] evaluating the role of oximes in OP poisoning were also identified. In addition, one prospective trial each on magnesium [31] and fresh frozen plasma therapy [32], two trials on haemoperfusion [33,34], as well as a Cochrane systematic review on the role of alkalinisation in human OP poisoning [35] were identified. The physiological basis of each of these therapies as available from animal and experimental data was also reviewed. Meta-analysis of trials was performed only on the eight studies that had a control arm (no oxime therapy) and a treatment arm (oxime therapy). Results of trials comparing different dose regimens of oximes were presented as such. It must be highlighted that a comparison among patients poisoned with varying OP compounds and treated with a range of oxime dosage schedules is difficult. The data presented below provides an 'overall' effect of oxime therapy in OP poisoning, and needs to be interpreted in the light of the limitations outlined above.

2.2.1 Oxime therapy

Data on mortality and need for mechanical ventilation was available in all the eight studies [15,17,19-22,25,26] that were included in the meta-analysis. Data on mechanical ventilation for one study was obtained through personal correspondence. The 209 patients with OP poisoning in the control arm received standard medical therapy including atropine, whereas 206 patients in the treatment arm received, in addition to standard therapy, atropine and either pralidoxime or obidoxime in varying doses. Different pralidoxime salts were used in the studies. Pralidoxime chloride was used in three studies [15,17,19]: iodide salt [18], pralidoxime methylsulphate [26] and pralidoxime sulfate [21] in one study each. The type of pralidoxime could not be ascertained in the remaining studies [20,22,25].

Table 1. Controlled trials on the role of oximes in organophosphate poisoning in humans: outcomes.

Trial	Year	Country	I/E	Type	Total number of patients		Number of patients who died		Mechanical ventilation		Intermediate syndrome	
					Rx	Control	Rx	Control	Rx	Control	Rx	Control
Duval [21]	1991	France	Y/N	Retrospective study	31	31	2	3	11	12	NA	NA
De Silva [19]	1992	Sri Lanka	Y/N	Historical controls	24	21	7	6	11	8	10	8
Abdollahi [20]	1995	Iran	Y/N	Retrospective study	17	17	3	3	7	8	NA	NA
Cherian [15]	1997	India	Y/Y	Randomised controlled trial*	55	55	16	3	37	22	36	19
Balali-Mood [26]	1998	Iran	Y/Y	Prospective non-randomised	20	43	6	4	6	2	NA	NA
Sungur [22]	2001	Turkey	N/N	Retrospective study	31	16	10	3	8	2	6	3
Cherian [17]	2005	India	Y/N	Randomised controlled trial*	10	11	1	1	7	4	NA	NA
Chugh [25]	2005	India	N/Y	Prospective non-randomised	15	15	1	0	6	5	NA	NA

*Randomised controlled trials had block randomization of size 4.

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I/E: Inclusion/exclusion criteria; N: No; NA: Not available; Rx: Oxime treated group; Y: Yes.

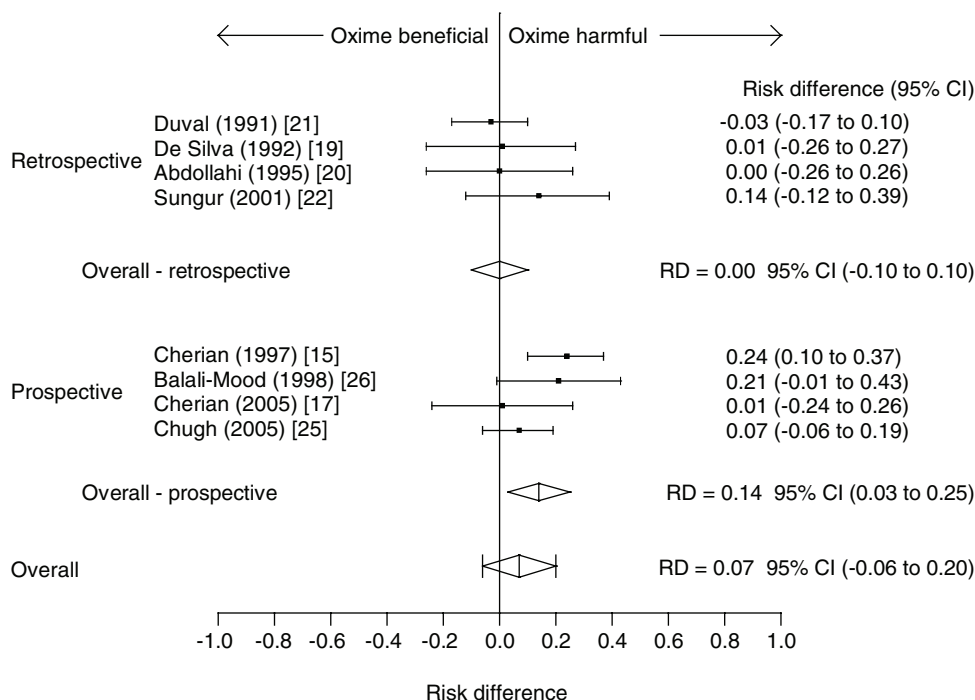


Figure 1. Association between oxime therapy and mortality. Forest plot representation using the random effects model depicting association between oxime therapy and mortality. The vertical straight line denotes null effect. The individual points denote the risk difference of each study, and the lines on either side the 95% CI. Combined estimates as well as individual estimates for the retrospective group, as well as the prospective study group are represented. Oxime therapy was not associated with a significant increase in mortality.

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RD: Risk difference.

The mortality rate was 11.0% (23 out of 209) in the control group and 20.7% (42 out of 203) in the treatment group. No association between oxime therapy and mortality was found (Figure 1) when all studies (retrospective and prospective) were pooled (RD = 0.07; 95% CI = -0.06 to 0.2; $p = 0.29$). Although there was a significant increase in mortality with oxime therapy in the prospective study cohort, no differences were observed in the retrospective cohort group. Although heterogeneity of treatment effects existed ($p = 0.07$) between the two study groups (retrospective and prospective), there was no heterogeneity within each specific group.

Invasive mechanical ventilation was required in 30.1% (63 out of 209 cases) in the control group and 45.8% (93 out of 203) in the treatment group receiving oxime therapy. Although the overall difference was not significant (RD = 0.15; 95% CI = -0.05 to 0.35; $p = 0.15$), oxime therapy was associated with significant increase in ventilatory support in the prospective cohort, but not in the retrospective cohort (Figure 2).

The need for intensive care was reported in all studies. In four studies, patients were admitted to the ICU only when indicated. However in the remaining four studies, all patients were admitted to the ICU for management. As this is not

likely to occur randomly, analysis was done only on the four studies where patient admission to the ICU was need-based. This analysis suggested a positive association (RD = 0.16; 95% CI = 0.02 – 0.31; $p = 0.03$) between oxime therapy and increased ICU admissions (Figure 3).

Two other RCTs [16,18] merit presentation and are summarised in Table 2. The first study [16], a randomised prospective trial that included 72 patients, was conducted in the early 1990s in South India, and compared 2 doses of oximes (a 1-g bolus versus a 12-g infusion over 3 days). In this study, higher oxime dose was associated with a trend to worse outcomes in terms of mortality (8 out of 36 versus 5 out of 36; $p = 0.35$), ventilatory requirements (24 out of 36 versus 17 out of 36; $p = 0.09$) and incidence of intermediate syndrome (20 out of 36 versus 13 out of 36; $p = 0.08$). However, a post-hoc analysis suggested a significant reduction in the incidence of an intermediate syndrome if at least 1 g of pralidoxime was administered within 12 h of intoxication [16].

In a more recent RCT from North India, Pawar *et al.* evaluated 200 moderately severe OP poisoned patients and demonstrated oxime effectiveness using very high doses of oximes (48 g of pralidoxime iodide in 48 h) in patients who presented early to hospital following OP poisoning [18].

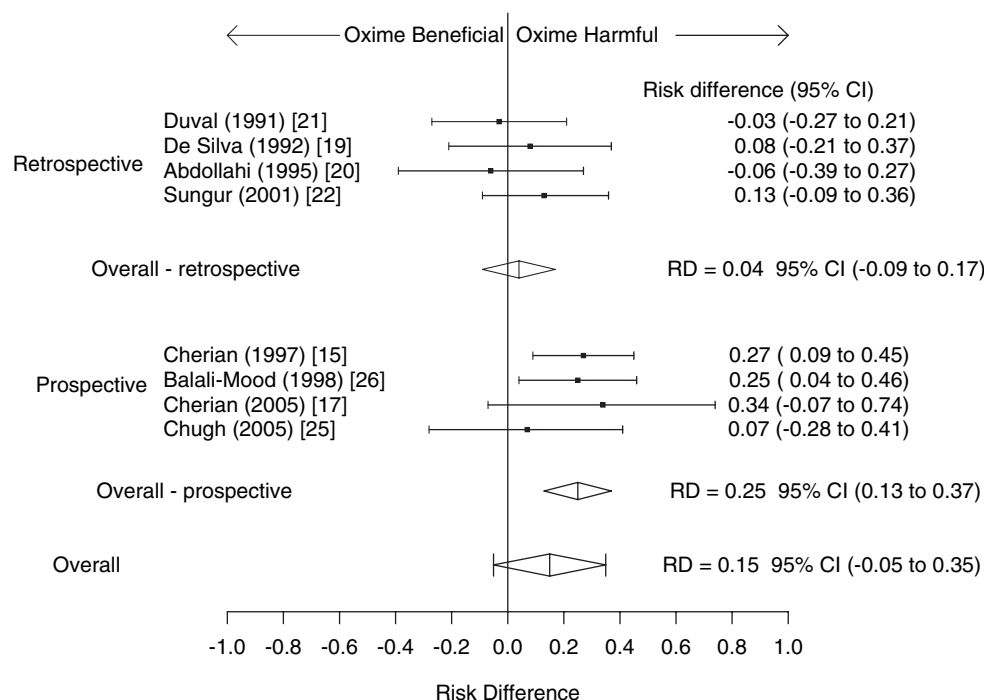


Figure 2. Association between oxime therapy and the need for ventilatory support. Association between oxime therapy and the need for ventilatory support is represented as risk difference using the random effects model. The vertical straight line denotes null effect. The individual points denote the risk difference of each study and the lines on either side the 95% CI. Individual estimates for the retrospective and prospective groups presented in addition to combined effects. Oxime therapy was not associated with a significant increase in ventilatory requirements.

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RD: Risk difference.

The control group that received 1 g of pralidoxime every 4 h for 48 h required higher doses of atropine compared with the treatment group (30 versus 6 mg; $p < 0.0001$), had higher intubation rates (88 versus 64%; $p = 0.0001$) and had more deaths (8 versus 1, $p = 0.04$). The authors concluded that high dose pralidoxime was associated with 'pronounced benefits' in patients with moderately severe OP poisoning [18].

2.2.2 Magnesium

One study evaluated the role of magnesium in human OP poisoning [31]. This study, reported from a single centre in Iran, included only patients who presented with oral ingestion of a known OP compound and who had not received advanced medical care in another medical centre. In this single-blinded, prospective, controlled trial, every fourth eligible patient was selected using systematic sampling to receive, in addition to pralidoxime or obidoxime, 4 g of magnesium sulphate over the first 24 h after admission. Although the mean daily oxime and atropine requirement was not statistically significant between the control and treatment groups, the mortality rate (14.7 versus 0%) and hospitalisation days (5.0 ± 0.82 versus 2.9 ± 0.66 days)

were significantly lower ($p < 0.01$) in those receiving magnesium (Table 3).

2.2.3 Haemoperfusion

Two recent publications presented data on the role of haemoperfusion in acute OP poisoning [33,34] (Table 3). In a retrospective report of 52 OP poisoned patients from Turkey, 25 patients with severe poisoning treated with haemoperfusion were compared with 27 patients with mild poisoning not receiving haemoperfusion [33]. Seven patients with severe poisoning died; all in the haemoperfusion group. Although cholinesterase (ChE) levels increased with haemoperfusion, both in survivors (from 1616.7 ± 612.3 mU/ml to 2169.0 ± 2837.8 mU/ml) and non-survivors (231.2 ± 148.7 mU/ml to 1860.4 ± 2137.9 mU/ml), the increment in ChE did not affect survival. These results contrasted with the prospective evaluation of 108 patients with severe dichlorvos poisoning from China; 67 received charcoal haemoperfusion and supportive therapy including pralidoxime (1–2 g every 6 h) whereas 41 patients received only supportive treatment [34]. Haemoperfusion was associated with significant benefits in mortality (7.5 versus 34.1%; $p < 0.01$), need for mechanical ventilatory support (13.4 versus 36.6%; $p < 0.01$), duration of ICU stay

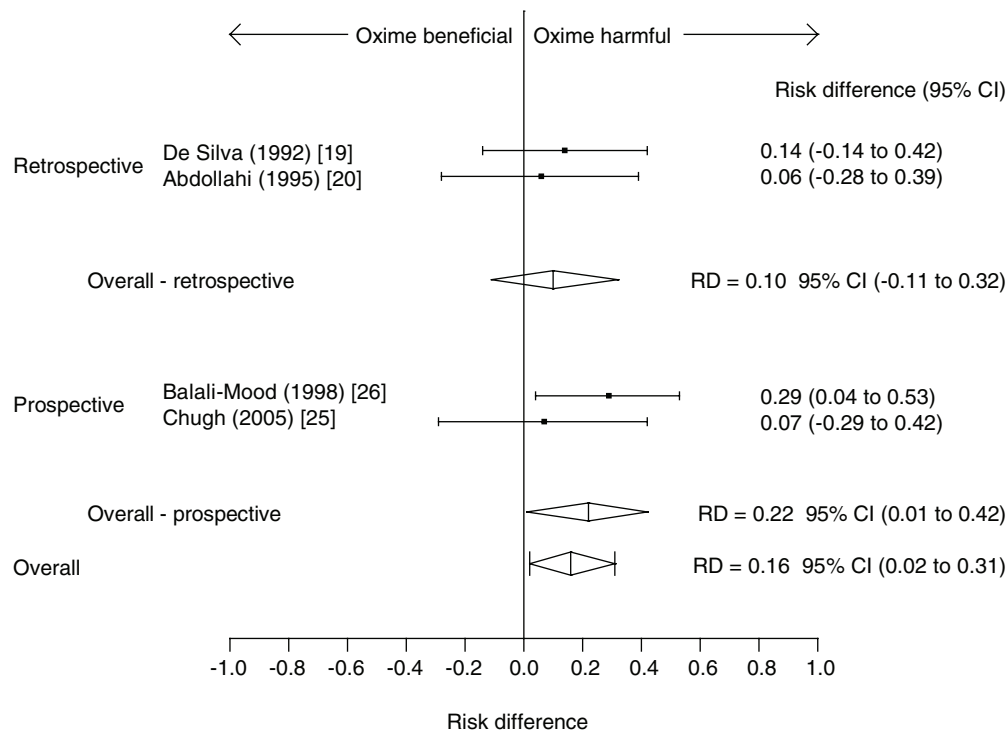


Figure 3. Association between oxime therapy and need for intensive care. Forest plot representation of the association between oxime therapy and need for intensive care therapy (random effects model). The vertical straight line denotes null effect. The individual points denote the risk difference of each study and the lines on either side the 95% CI. Oxime therapy is associated with an increased need for intensive care therapy.

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RD: Risk difference.

(4 ± 1.5 versus 6 ± 2 days, $p < 0.05$) and duration of coma (9 ± 6 versus 15 ± 8 h, $p < 0.01$).

2.2.4 Fresh frozen plasma

FFP as bioscavenger therapy has recently been suggested in the management of OP poisoning [32]. In a preliminary study of 33 OP poisoned patients from Turkey, 20 patients received atropine and pralidoxime and 9 patients in addition received 2 bags of plasma (300 – 400 ml/bag) daily from day 2 until they no longer required atropine [32]. Of the remaining four patients, two patients received plasma after onset of intermediate syndrome, one patient received atropine alone and the other patient received atropine and FFP. The authors observed that although intermediate syndrome occurred in 28.8% (6 out of 21) of patients who received atropine and pralidoxime, no patient developed intermediate syndrome if plasma was co-administered early (0 out of 9 cases). The mortality rate was 14.3% (3 out of 21) with atropine and pralidoxime. Although no deaths occurred in the subgroup of patients who received FFP prior to onset of intermediate syndrome ($n = 9$), the 2 patients who received FFP following onset of intermediate syndrome died (Table 3). Every 2 bags of plasma increased butyrylcholinesterase levels by 461.7 ± 142.1 IU/L.

2.2.5 Alkalinisation

In a controlled trial of 53 patients, in addition to standard treatment, 26 patients were randomised to receive sodium bicarbonate 5 mEq/kg in one h, followed by an infusion of 5 – 6 mEq/kg in 23 h [36]. The infusion was repeated daily till recovery or death. Although there was a trend towards less atropine use and shorter length of stay with bicarbonate therapy, no mortality benefits were observed (relative risk [RR] = 0.52; 95% CI = 0.05 – 5.39) [36].

2.3 Discussion

There has been renewed interest in the management of OP poisoned patients, fuelled not only by the large number of OP poisoned patients seen in the developing world, but also by acts of terrorism using nerve gas agents [37,38], as well as the constant threat of bioterrorism. Although there are likely to be several differences in the clinical profile and response to therapy between ingestional (suicidal) intoxication and inhalational (deliberate) exposure [39], it is important that all possible avenues of treatment are explored.

Although the importance of early management in terms of reduction of absorption of the toxin cannot be understated, at least with ingestional OP poisoning where patients present

Table 2. Prospective randomised controlled trials comparing two doses of pralidoxime in organophosphate poisoning.

Parameter	Study 1*			Study 2*		
	High dose	Low dose	p value	High dose	Low dose	p value
Number of patients	36	36		100	100	
Dose of oxime [‡]	12 g	1 g		48 g	12 g	
Age	24.9 (7.5) [§]	25.3 (10.8) [§]	0.89	28 (22 – 23) [¶]	29 (22 – 35) [¶]	NA
Time from ingestion to antidote administration	17.3 (11.6) [§]	16.8 (10.8) [§]	0.85	2 (1.5 – 2.4) [¶]	1.9 (1.0 – 2.5)	NA
Type of organophosphate compound						
Dimethyl organophosphate [#]	22 (61)	20 (56)		77 (77)	59 (59)	
Diethyl organophosphate [#]	8 (22)	9 (25)		23 (23)	41 (41)	
Unknown compound [#]	6 (17)	7 (19)		0	0	
Pseudocholinesterase levels, Mean (SD)**	441 (450)	339 (261)	0.24	NA	NA	NA
Median ~ (IQR) Butyrylcholinesterase levels**	NA	NA	NA	866 (752 – 939)	808 (535 – 911)	NA
Outcomes						
Duration of ICU stay (days)	6.8 (5) [§]	5.9 (4) [§]	0.55 ^{††}	NA	NA	NA
Number ventilated	24	17	0.09 ^{††}	64	88	0.0001 ^{††}
Duration of ventilation (days)	7.7 (5.2) [§]	6.8 (4.8) [§]	0.70 ^{††}	5 (4 – 5) [¶]	10 (10 – 12) [¶]	< 0.0001 ^{††}
Intermediate syndrome (n)	20	13	0.08 ^{††}	NA	NA	NA
Deaths (n)	8	5	0.35 ^{††}	1	8	0.04 ^{††}

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*Study 1 is [16] and study 2 [18].

[‡]In study 1, oxime dose over 72 h and in study 2, oxime dose over 48 h.

[§]Mean (SD).

[¶]Median (IQR 25 – 75%).

[#]Number (%) of patients.

**Normal pseudocholinesterase level: 3000 – 6000 IU/l; Normal butyryl cholinesterase levels: 2710 – 11510 IU/l.

^{††}p values from original paper.

ICU: Intensive care unit; IQR: Inter-quartile range; NA: Not available; SD: Standard deviation.

late in several centres, therapy is targeted at reversing the effects of the poison. In this context, oxime therapy has both clinical and economic implications. The clinical implications are in terms of oxime efficacy and the economic implications in terms of huge costs of stocking pralidoxime in every hospital in the US [40].

2.3.1 Oxime use in deliberate (inhalational) exposure

When evidence for oxime treatment in deliberate exposure (inhalational) is investigated, all we have are the reports from the Tokyo Sarin attack [37,38]. In the first report of the 640 patients presenting to the emergency department (ED), 528 patients were categorised as 'mild intoxication' and discharged from the ED [37]. A total of 107 patients (16.7%) with systemic symptoms and signs but not needing ventilatory support were categorised as moderate poisoning and 106 patients were administered 0 – 8 g of pralidoxime within 3 h of presentation; 85 patients (81.2%) required < 2 mg of atropine. There were only four patients with severe intoxication, one of whom died. One additional patient died in the ED. The quantity of atropine that was required (2 mg) in this cohort of 'moderate poisoning' was much smaller than

doses required (23.4 mg) during initial resuscitation in 'severe' ingestional poisoning [41] and this suggests that the 102 patients with moderate poisoning may have had only mild intoxication.

Nozaki *et al.*, in a separate cohort, reported 113 patients presenting with exposure to sarin [38]. Of these cases, 1 was dead on arrival and only 15 patients needed admission for further treatment, suggesting that only a very small proportion of patients presented with significant toxicity. Only 1 g of pralidoxime was used in this case-report.

Newmark reviewing and translating Farsi-language articles published by Dr Syed Abbas Foroutan, describes Foroutan's first hand accounts of battle-field nerve-agent casualty care, and compares his recommendations with US/North Atlantic Treaty Organisation (NATO) doctrines [42]. Foroutan stated that pralidoxime was '*totally ineffective*' against tabun and soman, but in patients who received obidoxime, '*reaction was generally immediate and complete*'. He also suggested that in soman poisoning, obidoxime use was limited by the formation of potent inhibitors such as phosphonyl oximes. Newmark referred to the short '*ageing*' half-life of sarin (5 h) and the assumed lack of efficacy of oximes in Foroutan's reports as

Table 3. Summary of adjuncts and alternative therapies to oximes for human organophosphate poisoning.

Intervention (Treatment)	Study Author (Year)	Type of study	Number of Patients		Mortality		Mechanical ventilation		Intermediate syndrome	
			Rx ^a arm	Control arm	Rx ^a arm	Control arm	Rx ^a arm	Control arm	Rx ^a arm	Control arm
Magnesium	Pajoumand [31] (2004)	Prospective	11	34	0 (0%)	5 (14.7%)	NA	NA	NA	NA
Haemoperfusion	Peng [34] (2004)	Controlled	67	41	5 (7.5%)	14 (34.1%)	9 (13.4%)	15 (36.6%)	20 (29.9%)	15 (36.6%)
	Altintop [33] (2005)	Retrospective	25	27	7 (28%)	0 (0%)	NA	NA	NA	NA
Fresh frozen plasma (FFP)	Güven [32] (2004)	Prospective partially randomised	11	22	2* (18.2%)	4* (18.2%)	NA	NA	0† (0%)	6 (28.6%)
Alkalinisation	Roberts [35] (2005)	Randomised trial (Systematic review)	26	27	1 (3.8%)	2 (7.4%)	NA	NA	NA	NA

*The 2 deaths in the treatment arm occurred in the 2 patients receiving FFP following onset of intermediate syndrome; the 4 deaths in the control arm include 3 deaths in patients treated with atropine and pralidoxime and 1 treated with atropine alone.

†No patient developed intermediate syndrome following FFP administration (0/9), although 2 patients received FFP after onset of intermediate syndrome.

All data are presented as number of patients for mortality, mechanical ventilation and intermediate syndrome. Values in parenthesis indicate percentages.

NA: Data not available from study; Rx^a: Treatment arm receiving the intervention listed in column 1.

being due to delays in transfer to treatment units [42]. However, the US/NATO doctrine mandates that each soldier carries 3 doses each of atropine (2 mg each) and pralidoxime (600 mg each) in addition to 10 mg of diazepam, and that 1800–2000 mg of pralidoxime be administered intra-muscularly or intravenously post-exposure.

It is difficult to ascertain oxime benefit from these reports, although it makes physiological sense to administer the antidote in circumstances described above, where the access to the antidote may be quick. The dose of antidote again is not clear, although 1–2 g of pralidoxime administered immediately post-exposure appears to be the norm.

2.3.2 Oxime use in suicidal (ingestional) exposure

There is considerably more information on oxime use in ingestional exposure, although the methodology of some of these trials has been questioned in systematic reviews [6,30]. The evidence presented in the present meta-analysis, as well as 2 other previous meta-analyses [5,28], suggest that oxime therapy may be associated with adverse effects when used in ingestional OP poisoning. These effects are seen particularly in the current meta-analysis for need for intensive care therapy. In the prospective trials group, there was an association between oxime therapy and increased mortality and need for invasive mechanical ventilation. However, overall effects are insignificant. The recent publication on high dose oxime therapy appears to contradict the results of the meta-analysis in that a significant improvement in both mortality and need for ventilation was demonstrated; albeit this trial did not have a placebo arm [18].

On closer examination, the pivotal factor determining response to oxime therapy appears to be the time of administration of the antidote. The studies included in the meta-analysis reported varied times of presentation up to 24 h, and so it was not possible to derive any meaningful time-effect of antidote administration.

However the two RCTs comparing two treatment regimens of pralidoxime offer some insight and suggest that pralidoxime may be effective if administered sufficiently early [16,18]. In the study by Johnson *et al.*, although overall effects were adverse with higher doses of pralidoxime (12 g) compared with single dose (1 g), a post-hoc analysis suggested a reduction in the incidence of intermediate syndrome if at least 1 g of pralidoxime was administered within 12 h of intoxication [16]. In the more recent study by Pawar *et al.*, it appeared that a majority of patients presented within 2.5 h of ingestion and very high doses (48 g in 48 h) of pralidoxime iodide conferred outcome benefit [18]. The iodide salt used in this trial is considered to be less potent (by a factor of 1.5) than pralidoxime chloride [43]. However, this dose still works out to be higher (22.2 mg/kg/h of pralidoxime iodide or 14.5 mg/kg/h of pralidoxime chloride) for a 45-kg Indian patient [6] than the WHO-sponsored dose of 8 mg/kg/h of pralidoxime chloride [6]. This trial did not address the role of oximes in severe intoxications or late presentations

of OP poisoning. It was again unclear from the trial [18] if such high doses of oximes would be sufficient in mega-dose intoxications, where even with optimal plasma concentrations, oximes may not be able to cope with the fast re-inhibition of re-activated AChE [5].

Apart from (limited) clinical data, there is also electrophysiological evidence that the time of administration of antidote may determine response. Electrophysiological studies in 6 patients presenting with acute OP poisoning demonstrated electrophysiological improvements only when the antidote was administered within 12 h of poisoning. These improvements were not observed if the antidote was administered after 26 h [44]. In the same study, obidoxime worsened neuromuscular blockade when administered 66 h after intoxication [44]. In a subsequent study, obidoxime was ineffective in oxydemeton-methyl poisoning when the time elapsed between ingestion and therapy was longer than one-day [45]. When obidoxime was administered shortly after ingestion (1 h), reactivation was nearly complete [45].

More recently Khan *et al.*, studying the temporal profile of butyrylcholinesterase (BuChE) and *in vitro* pralidoxime-reactivated BuChE in a cohort of 25 OP poisoned patients, observed that the reactivation potentials of BuChE (the difference between oxime-reactivated and unreactivated enzyme activity) declined significantly with time after OP poisoning [46]. They also observed that the reactivation potential of the enzyme at admission decreased significantly with increasing severity of poisoning.

These clinical, biochemical and electrophysiological studies suggest that oxime therapy may be beneficial in (very) early presentations of OP poisoning. Determining the optimal period of time that oximes could be administered following intoxication is a difficult question that has not been answered in the trials so far. This might well depend on the propensity of the OP compound to induce ageing as well as the metabolic transformation of the OP compound to toxic metabolites.

It has been recognised that phosphorylated AChE is prone to 'ageing', a process by which one alkyl side-chain of the phosphoryl moiety is removed non-enzymatically, leaving a hydroxyl group in its place [5]. Once AChE is 'aged', regeneration is not possible except by *de novo* synthesis of the enzyme. Ageing and regeneration of the AChE occurs concomitantly with ongoing inhibition of AChE by the OP compound, with the portion of the aged enzyme being dependent on the presence of excess inhibitor (in this instance, the OP compound). Therefore, the therapeutic window for oximes is very much determined by the rate of ageing, ongoing inhibition as well as spontaneous reactivation of phosphorylated AChE. We also know that the ageing and reactivation kinetics are different for the dimethyl and the diethyl phosphoryl compound. Early *in vitro* studies by Reiner and Plestina suggested spontaneous *in vitro* reactivation rates of 1.3 and 2.1 h for dimethyl and diethyl phosphorylated AChE, respectively [47]. More recently, the half-life of ageing

of dimethyl and diethyl phosphorylated AChE, as determined in isolated human red cells *in vitro*, were 3.7 and 33 h, respectively [48-50]. If we then take the potential therapeutic window for the administration of the antidote as four times the half-life [6], antidotes may be beneficial for a maximum of 13 h for the dimethyl phosphoryl group and 132 h in the diethyl phosphoryl group. It has been suggested that in intoxications with a dimethyl phosphoryl insecticide, virtually all AChE will be in the aged form in one day, and that oxime therapy will be useless by that time [51]. Although the 'therapeutic windows' suggested above are theoretically plausible, it must be highlighted that calculations based only on 'ageing' may be inaccurate, as the rate of ageing depends not only on the pharmacokinetics of the OP compound, but also on the presence of excess inhibitor (OP) that might further inhibit regenerated enzyme. Thus, the ideal time frame for antidote administration needs to be established by clinical trials.

2.3.3 Adjunct therapy

The one trial on magnesium suggested a possible role in OP poisoning. Although small sample size and lack of placebo administration in the control arm limit inferences from this trial [31], there is some physiological basis for magnesium use. Magnesium acts on the heart by suppressing the early-after depolarisations and the triggered activity [52,53]. This may effect a reduction in the occurrence of the arrhythmias. There is also evidence from animal experiments that magnesium reverses the decrement in the force of contraction and compound muscle action potential (CMAP) in the rat diaphragm muscle treated with diisopropyl fluorophosphate [54]. Given cheaper costs and its low toxicity profile, magnesium appears worthy of further study.

The two studies on haemoperfusion appear to give us conflicting results. Although Peng *et al.* [34], in the prospective evaluation, demonstrated benefits with haemoperfusion, Altintop *et al.* [33] found more deaths in a retrospective cohort that underwent haemoperfusion. However, the crucial factor determining response, again as with oximes, may be the time of provision of haemoperfusion. The time from intoxication to provision of haemoperfusion was only 5.6 ± 4.8 h in the study by Peng *et al.* [34] and significantly different (t-test p value = 0.003) to the 27 ± 32 h in the Altintop *et al.* study [33]. In an earlier case-report, combined haemodialysis and haemoperfusion initiated within 2 h of severe dimethoate poisoning reduced serum dimethoate levels from 3110 ng/ml to undetectable after 18 h of extracorporeal therapy [55]. These observations argue haemoperfusion effectiveness in early OP presentations when tissue distribution would be incomplete. However, the large volume of distribution of some of the OP compounds may limit benefit [33].

Bioscavenger therapy, a novel modality of treatment with the use of ChEs such as B-esterases [56], has been recently evaluated in human OP poisoning using FFP [27] as substrate for OPs or anti-cholinesterase agents. This approach may be

considered a paradigm shift with the cholinesterase being used as an 'anti-OP' agent rather than the OP compound acting as an 'anti-cholinesterase' or cholinesterase inhibitor [56]. In the pilot study described above [32], clinical benefits were observed; albeit interpretations limited by lack of allocation concealment and 'partial' randomisation. The initial postulate that ChEs may be the 'secret ingredient' has been questioned [57], with the observations that albumin may also serve to scavenge OP [58]. Furthermore, the proportionately small quantities of BuChE in transfused FFP compared with poison-loads, as well as late FFP administration (in the Guven [32] study), suggest that the beneficial effect of FFP might actually be related to hydrolysis of acetylcholine in the blood rather than primary scavenging of free OP. Although the relative ease of availability of FFP (as compared with recombinant or pooled cholinesterase) in developing countries may spur research in these countries, issues such as transfusion-related problems might preclude widespread acceptance of this treatment.

The systematic review on alkalinisation in OP poisoning has suggested that there was insufficient evidence from the single RCT and the four excluded uncontrolled trials to recommend alkalinisation in OP poisoning [35].

2.4 Conclusion

It is evident from the above trials that considerable progress has been made in the last decade in the understanding and management of OP poisoning. It is also clear that OP poisoning encompasses a wide variety of compounds with different pharmacokinetic and ageing characteristics. The distinction between diethyl and dimethyl OP poisoning and the differential ageing and re-activation kinetics suggest that dimethyl OP poisoning may be more refractory to oxime therapy [49]. Furthermore, it appears that the time of antidote administration may be the single most important factor that affects outcomes. Based on the study of Pawar *et al.* that suggested a treatment-effect of high dose pralidoxime (iodide) in very early presentations of 'moderately severe' OP poisoning, oxime therapy may be considered in this select group of patients [18]. However, the treating physician should continue to use judgement regarding oxime use in severe intoxications as well as late presentations of OP poisoning where the evidence for benefit is not demonstrated. In such patients, oxime therapy may be associated with harm, given the propensity of the OP compound to induce ageing as well as form phosphoryl oximes that have high anticholinergic activity [59]. Newer modalities of treatment and adjunct therapies offer hope in the drive to reduce the mortality burden with OP poisoning.

3. Expert opinion

OP poisoning can be considered an epidemic in several developing countries. A multi-faceted campaign is urgently required to arrest this epidemic. The first step towards this appears to be the phasing out of the highly toxic insecticides.

This strategy, implemented in Sri Lanka in a phased manner over nearly 20 years, with the banning of WHO Class-I OP compounds like monocrotophos and methamidophos, as well as other carbamates and organochlorines, has led to a fall in the case fatality proportion [60]. Countries such as India that do not have such legislation in place continue to encounter severe intoxications with the highly toxic insecticides [61]. With the recent advent of less toxic but equally potent insecticides, it appears to be a good strategy to phase out the more toxic compounds and replace these with compounds such as pyrethroids and neo-nicotinoids, that not only have versatile applications, but also favourable environmental and toxicological profiles [62]. However, phasing out the toxic insecticides may be met with resistance not only from the flourishing industry, but also from the consumer, who might end up spending more for the newer compounds. Thus, such a strategy would require a strong legislative stand by the government. Additional factors that might halt this epidemic would be community education and safe pesticide storage [102], as well as dispensing diluted OP compounds. Legislation to dispense only dilute OP compounds may be considered analogous to blister-packaging and limiting the size of packets of paracetamol and salicylates sold over the counter; a strategy in the UK that resulted in substantial beneficial effects on mortality and morbidity [63]. Although such strategies might reduce the severity of poisoning, it would also increase packaging, transport and storage costs that would probably be passed on to the consumer.

What about the management of patients presenting with OP poisoning? The results of the earlier oxime trials and the meta-analyses did not demonstrate a favourable association between oxime use and outcomes, albeit the most recent trial suggested oxime benefit in very early presentations of moderately severe OP poisoning. Given that only ~ 15% of OP poisoned patients present to our hospital within 6 h of intoxication [16], and the fact that this is likely to be experience

of other centres as well, it is probable that late presenters with severe intoxications may not respond favourably to oxime therapy. It is to be emphasised that in the absence of strong evidence for oxime benefit in OP poisoning, textbooks still recommend the routine use of oximes for 'all' OP poisoning. In the era of evidence-based practice of medicine, we caution against 'indiscriminate' oxime use in 'all' patients, given the reasons and therapeutic limitations outlined in this review. A further caution to enthusiastic oxime use is the fact that oximes by themselves can produce respiratory (inhibition of patient's respiration), cardiac (arrhythmias, atrioventricular block, asystole) and neurological (weakness) manifestations [5]. The role of good supportive therapy and improved standards of care in the developing countries, suggested in recent correspondence [64], may be more crucial in determining successful outcomes in OP poisoning rather than costly therapies that are still debatable [65]. Our understanding of the role of antidote therapy will also improve only with properly conducted randomised trials that are stratified appropriately for the type of OP compound (dimethyl versus diethyl), severity of intoxication (mild, moderate, severe) as well as time of presentation (early versus late). It must be emphasised again that we still do not have accurate criteria to categorise patients as mild, moderate, severe or very severe intoxications, and this would be crucial to assess effectiveness of therapy in subgroups where differential treatment effects may be observed in the distinct categories of severity.

OP poisoning could also be viewed from a different perspective. Rather than focussing on a single modality of treatment (e.g., oximes), a multi-modal approach targeting several aspects of poisoning, including reduction of absorption of the poison, neutralising the toxin as well as its effects and enhancing elimination, might be the answer to improving outcomes in OP poisoning. With work on several adjuncts in progress, we are likely to see positive strides in this march against insecticide poisoning.

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